

OMNIX QUANTUM -- Pressure Evidence Packet

TA-14 Third-Pass: Executable Pressure-Evidence Scenarios

RFC-ATF-1 through RFC-ATF-6 | ADR-157 . ADR-159 . ADR-200 | 2026-05-27 21:56:26 UTC

Packet Identity

Packet ID	OMNIX-PEP-73F019BA3CB0411E
Packet Type	PRESSURE_EVIDENCE_PACKET
OMNIX Version	OMNIX-2.6.0
TA-14 Target	Third-Pass Executable Pressure-Evidence Review
PQC Algorithm	ML-DSA-65 (Dilithium-3, FIPS 204)
PQC Standard	NIST FIPS 204 -- NIST Level 3
Generated	2026-05-27 21:56:26 UTC
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Executive Summary

This Pressure Evidence Packet demonstrates OMNIX ATF behavior under three adverse runtime conditions relevant to the TA-14 Third-Pass review. Each scenario executes a complete 8-step admissibility chain (Source State -> Record -> Continuity -> Admissibility -> Binding -> Commit -> Execution -> Outcome) and produces PQC-signed artifacts verifiable offline without OMNIX runtime access. The ephemeral ML-DSA-65 keypair embedded in this packet covers all three scenarios. Verification: 69/69 checks PASS.

OFFLINE VERIFICATION: 69/69 PASS -- 0 FAIL

Pressure Scenarios -- Overview

Scenario	Adverse Condition	System Response	TA-14 Finding	Invariants
P1 Authority Drift	DR valid at record time, expired before admissibility gate (2s TTL, 3s delay)	TAR REJECTED Binding REFUSED Hard Refusal issued	DRIFT PROVEN -- temporal gate blocks expired authority	TAR-INV-003 TAR-INV-006 ATF-INV-001
P2 Stale Continuity	CES temporal health degraded to 3% (DR near-expiry)	HALT posture raised Binding REFUSED Hard Refusal issued	STALENESS PROVEN -- continuity gate blocks degraded CES	RGC-INV-002 RGC-INV-007 RGC-INV-003
P3 RC TTL Enforcement	Reauthorization Challenge expired 5 min before commit gate	Binding HALT Commit BLOCKED Sub-tasks revoked	RC EXPIRY PROVEN -- auto-HALT + sub-task revocation enforced	RGC-INV-008 RGC-INV-003 ATF-INV-001

Invariants Stress-Tested

- * ATF-INV-001: Authority never expands through delegation (MAR) -- verified across all 3 scenarios
- * TAR-INV-003: Expired/exhausted DRs produce admission_status=REJECTED -- P1
- * TAR-INV-006: DR remaining validity window enforced at admission -- P1
- * RGC-INV-002: CES computed from real-time values only -- P2
- * RGC-INV-003: HALT terminates execution and revokes sub-tasks -- P3
- * RGC-INV-007: CES inputs must meet freshness requirements -- P2
- * RGC-INV-008: RC TTL enforced -- auto-HALT on expiry -- P3

P1 -- Authority Drift

DR valid at record time, expired before admissibility gate

Step	Value	Detail
1. Source State	Captured	request + pressure_note recorded
2. Record (DR)	ATFDR-1E1E0B95FBC44B29	budget=75.0/100.0 pqc=ML-DSA-65
3. Continuity	RCR issued	ces_score=N/A band=N/A
4. Admissibility	TAR: REJECTED	DR expired at nanosecond 1779918979566164992; execution at n
5. Binding	REFUSED	DR expired at nanosecond 1779918979566164992; execution at n
6. Commit	BLOCKED	Binding refused -- authority expired during transit (drift)
7. Execution	HARD_REFUSAL	REFUSAL-DBD8739BD09048BE
8. Outcome	AUTHORITY_DRIFT_REFUSAL	execution_occurred=False

TA-14 Finding: DRIFT PROVEN -- temporal admissibility gate blocks expired authority before binding

P2 -- Stale Continuity

CES temporal health degraded to HALT/WARNING band

Step	Value	Detail
1. Source State	Captured	request + pressure_note recorded
2. Record (DR)	ATFDR-C782C925B35A46B9	budget=75.0/100.0 pqc=ML-DSA-65
3. Continuity	RCR issued	ces_score=63.4 band=MONITORING
4. Admissibility	TAR: ADMITTED	admitted
5. Binding	REFUSED	HALT posture -- CES=63.4 (MONITORING) -- temporal health=3.0%
6. Commit	BLOCKED	HALT posture propagated from continuity -- CES=63.4
7. Execution	HARD_REFUSAL	REFUSAL-24E63155E47540B4
8. Outcome	STALE_CONTINUITY_REFUSAL	execution_occurred=False

TA-14 Finding: STALENESS PROVEN -- continuity gate detects degraded health and blocks execution

P3 -- RC TTL Enforcement

Reauthorization Challenge expired mid-session -> HALT

Step	Value	Detail
1. Source State	Captured	request + pressure_note recorded
2. Record (DR)	ATFDR-626222C28CFD4DCF	budget=75.0/100.0 pqc=ML-DSA-65
3. Continuity	RCR issued	ces_score=92.47 band=NOMINAL
4. Admissibility	TAR: ADMITTED	admitted
5. Binding	HALT	RC TTL expired -- RGC-INV-008: auto-HALT on RC expiry
6. Commit	BLOCKED	HALT propagated from RC TTL expiry -- RGC-INV-003
7. Execution	HALT	HALT-63DABEDE2428449D
8. Outcome	RC_TTL_HALT	execution_occurred=False

TA-14 Finding: RC EXPIRY PROVEN -- auto-HALT blocks execution and revokes sub-tasks

Offline Verification Report

Verification performed offline using only the ephemeral ML-DSA-65 public key embedded in the packet. No OMNIX runtime, database, or network access required. Each artifact's content_hash and PQC signature are independently verified.

VERDICT: ALL 69 VERIFICATIONS PASS

Passed: 69 | Failed: 0 | Skipped: 0 | Verified: 2026-05-27 21:56:35 UTC

P1 -- Authority Drift -- 16/16 PASS

[PASS] P1_AUTHORITY_DRIFT.DR.MAR	granted=75.0 delegator=100.0
[PASS] P1_AUTHORITY_DRIFT.DR.HASH	stored=c9b617bb5cd3b89d... expected=c9b617bb5cd3b8
[PASS] P1_AUTHORITY_DRIFT.DR.SIG	OK
[PASS] P1_AUTHORITY_DRIFT.TAR.HASH	stored=553e0aebc4f61cfc... expected=553e0aebc4f61c
[PASS] P1_AUTHORITY_DRIFT.TAR.STATUS	status=REJECTED
[PASS] P1_AUTHORITY_DRIFT.TAR.SIG	OK
[PASS] P1_AUTHORITY_DRIFT.RCR.HASH	stored=15644680f9759a73... expected=15644680f9759a
[PASS] P1_AUTHORITY_DRIFT.RCR.SIG	OK
[PASS] P1_AUTHORITY_DRIFT.BIND.HASH	stored=b7c3c4a4cd98809a... expected=b7c3c4a4cd9880
[PASS] P1_AUTHORITY_DRIFT.BIND.STATUS	status=REFUSED
[PASS] P1_AUTHORITY_DRIFT.BIND.SIG	OK
[PASS] P1_AUTHORITY_DRIFT.EXEC.HASH	stored=726f7fb68a5c677b... expected=726f7fb68a5c67
[PASS] P1_AUTHORITY_DRIFT.EXEC.TYPE	type=HARD_REFUSAL
[PASS] P1_AUTHORITY_DRIFT.EXEC.SIG	OK
[PASS] P1_AUTHORITY_DRIFT.OUT.HASH	stored=1d13b1f78b5122ab... expected=1d13b1f78b5122
[PASS] P1_AUTHORITY_DRIFT.OUT.EXEC	

P2 -- Stale Continuity -- 16/16 PASS

[PASS] P2_STALE_CONTINUITY.DR.MAR	granted=75.0 delegator=100.0
[PASS] P2_STALE_CONTINUITY.DR.HASH	stored=9a2899c9e272176a... expected=9a2899c9e27217
[PASS] P2_STALE_CONTINUITY.DR.SIG	OK
[PASS] P2_STALE_CONTINUITY.TAR.HASH	stored=1c861d42643d2fb3... expected=1c861d42643d2f
[PASS] P2_STALE_CONTINUITY.TAR.STATUS	status=ADMITTED
[PASS] P2_STALE_CONTINUITY.TAR.SIG	OK
[PASS] P2_STALE_CONTINUITY.RCR.HASH	stored=9d80dad56b757ceb... expected=9d80dad56b757c
[PASS] P2_STALE_CONTINUITY.RCR.SIG	OK
[PASS] P2_STALE_CONTINUITY.BIND.HASH	stored=1d7b62cc5456e1d5... expected=1d7b62cc5456e1
[PASS] P2_STALE_CONTINUITY.BIND.STATUS	status=REFUSED
[PASS] P2_STALE_CONTINUITY.BIND.SIG	OK
[PASS] P2_STALE_CONTINUITY.EXEC.HASH	stored=83dad4f34ef8ae9... expected=83dad4f34ef8a
[PASS] P2_STALE_CONTINUITY.EXEC.TYPE	type=HARD_REFUSAL
[PASS] P2_STALE_CONTINUITY.EXEC.SIG	OK
[PASS] P2_STALE_CONTINUITY.OUT.HASH	stored=e60fb50d7b5f572a... expected=e60fb50d7b5f57
[PASS] P2_STALE_CONTINUITY.OUT.EXEC	

P3 -- RC TTL Enforcement -- 16/16 PASS

[PASS] P3_RC_TTL_ENFORCEMENT.DR.MAR	granted=75.0 delegator=100.0
[PASS] P3_RC_TTL_ENFORCEMENT.DR.HASH	stored=7d540f7788a141ac... expected=7d540f7788a141
[PASS] P3_RC_TTL_ENFORCEMENT.DR.SIG	OK
[PASS] P3_RC_TTL_ENFORCEMENT.TAR.HASH	stored=61ce395dfc5f076e... expected=61ce395dfc5f07

```
[PASS] P3_RC_TTL_ENFORCEMENT.TAR.STATUS      status=ADMITTED
[PASS] P3_RC_TTL_ENFORCEMENT.TAR.SIG          OK
[PASS] P3_RC_TTL_ENFORCEMENT.RCR.HASH         stored=e89eaf72ac5b4fe1... expected=e89eaf72ac5b4f
[PASS] P3_RC_TTL_ENFORCEMENT.RCR.SIG          OK
[PASS] P3_RC_TTL_ENFORCEMENT.BIND.HASH        stored=639f669040da0a5d... expected=639f669040da0a
[PASS] P3_RC_TTL_ENFORCEMENT.BIND.STATUS      status=HALT
[PASS] P3_RC_TTL_ENFORCEMENT.BIND.SIG          OK
[PASS] P3_RC_TTL_ENFORCEMENT.EXEC.HASH        stored=45fd8590fe4a671c... expected=45fd8590fe4a67
[PASS] P3_RC_TTL_ENFORCEMENT.EXEC.TYPE        type=HALT
[PASS] P3_RC_TTL_ENFORCEMENT.EXEC.SIG          OK
[PASS] P3_RC_TTL_ENFORCEMENT.OUT.HASH         stored=022a4f470f6b6850... expected=022a4f470f6b68
[PASS] P3_RC_TTL_ENFORCEMENT.OUT.EXEC
```

P1 -- Pressure Assertions -- 5/5 PASS

```
[PASS] P1.ASSERT.TAR_REJECTED      status=REJECTED
[PASS] P1.ASSERT.BIND_REFUSED      status=REFUSED
[PASS] P1.ASSERT.DRIFT
[PASS] P1.ASSERT.NO_EXEC
[PASS] P1.ASSERT.DRIFT_NOTE
```

P2 -- Pressure Assertions -- 5/5 PASS

```
[PASS] P2.ASSERT.CES_HALT          ces=63.4 band=MONITORING
[PASS] P2.ASSERT.STALE
[PASS] P2.ASSERT.HALT_POSTURE
[PASS] P2.ASSERT.BIND_REFUSED      status=REFUSED
[PASS] P2.ASSERT.NO_EXEC
```

P3 -- Pressure Assertions -- 6/6 PASS

```
[PASS] P3.ASSERT.RC_EXPIRED
[PASS] P3.ASSERT.HALT_RC
[PASS] P3.ASSERT.BIND_HALT         status=HALT
[PASS] P3.ASSERT.SUBTASKS
[PASS] P3.ASSERT.HALT_TYPE         type=HALT
[PASS] P3.ASSERT.NO_EXEC
```

Packet-Level Assertions -- 5/5 PASS

```
[PASS] PKT.VERSION                 version=1.0.0
[PASS] PKT.PQC_KEY
[PASS] PKT.SCENARIOS               found=3
[PASS] PKT.INVARIANTS
[PASS] PKT.VERIF_INSTR
```

Verifier Scope

- * PQC keypair consistency across all scenarios (same ephemeral key)
- * DR content_hash + PQC signature for all 3 scenarios
- * TAR content_hash + admission_status for all 3 scenarios
- * Continuity record hash + PQC signature for all 3 scenarios
- * Binding record hash + PQC signature for all 3 scenarios
- * Refusal/HALT receipt hash + PQC signature for all 3 scenarios
- * Outcome hash integrity for all 3 scenarios
- * Pressure invariant assertions (drift confirmed, staleness confirmed, RC expired)
- * execution_occurred=False for all 3 pressure scenarios

Scope limits (by design -- offline packet):

- * Does NOT verify live database state or Railway deployment
- * Does NOT verify external market data in source_state
- * Does NOT require OMNIX runtime, network, or API access
- * Governance policy parameters documented in RFC-ATF-1 through RFC-ATF-6 (Zenodo/Figshare DOIs)

Governance Policy Parameters -- RFC Reference Map

The following maps TA-14 governance policy parameters to the RFC-ATF specifications that define them. Each parameter is covered by a published RFC with a permanent DOI.

TA-14 Parameter	OMNIX Value / Constraint	RFC Reference	DOI
Authority ceiling (max delegation budget)	0.0 <= granted <= delegator (MAR invariant -- ATF-INV-001)	RFC-ATF-1 ss4.2	10.5281/zenodo.20155016
Authority expiry (DR TTL enforcement)	Checked at nanosecond precision at each admissibility gate (TAR-INV-006)	RFC-ATF-2 ss3.1	10.5281/zenodo.20241344
Continuity health (CES formula)	CES = (T*0.30)+(B*0.30)+(D*0.20)+(I*0.20)	RFC-ATF-2 ss4	10.5281/zenodo.20241344
Freshness requirement (staleness bound)	CES inputs computed from real-time values only (RGC-INV-007)	RFC-ATF-2 ss4.3	10.5281/zenodo.20241344
RC TTL (reauth challenge)	Auto-HALT on RC expiry Sub-tasks revoked (RGC-INV-008)	RFC-ATF-2 ss5	10.5281/zenodo.20241344
HALT propagation	HALT terminates execution and revokes sub-tasks (RGC-INV-003)	RFC-ATF-2 ss5.1	10.5281/zenodo.20241344
Risk ceiling (mandate enforcement)	RISK-MANDATE-2026-Q2 risk_class=HIGH	RFC-ATF-3 ss3	10.5281/zenodo.20247342
Mandate binding (MBR certification)	MANDATE-BOUND / MANDATE-ALIGNED / UNCERTIFIED	RFC-ATF-4 ss4	10.5281/zenodo.20368895
PQC signing (receipt integrity)	ML-DSA-65 (Dilithium3, NIST SP 800-09) NIST Level 3 -- all receipts signed	RFC-ATF-1 ss6 / RFC-ATF-6 ss3	10.5281/zenodo.20155016 10.5281/zenodo.20393088

Offline Verification Command

Run the following command against the companion JSON packet:

```
$ python scripts/verify_pressure_evidence_packet.py omnix_pressure_evidence_packet_20260527_215626.json
```

No OMNIX runtime, database, or network access required. The verifier uses only the ML-DSA-65 public key embedded in the packet. Exit code 0 = all checks pass. Exit code 1 = one or more checks fail.